

Competition-Based Stochastic Learning Particle Swarm Optimization Algorithm for Large-Scale Optimization Problem

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Abstract—With the development of information technology, large-scale optimization problem is the core issue in various applications. Due to the high dimension characteristics, large-scale optimization problems are challenging to solve. Therefore, this paper proposes a competition-based stochastic learning particle swarm optimization algorithm (CSLPSO) for solving large-scale optimization problems efficiently, which has two novel designs. First, a competition-based stochastic learning strategy is proposed to improve the exploration ability of PSO. Second, a parameter control mechanism suitable for the CSLPSO algorithm is proposed, so that the algorithm does not require manual parameter selection. In the experiment, the CSLPSO algorithm is applied to 12 large-scale problems with state-of-the-art algorithms as competitors. The experimental results show the effectiveness of the proposed CSLPSO.

Keywords—particle swarm optimization (PSO), competition-based stochastic learning strategy (CSL), large-scale optimization problems, parameter control mechanism

I. INTRODUCTION

Optimization problems have long been a fundamental research topic due to their obvious potential for applications in almost all real-world systems, such as feature selection [1], recommendation system [2], wireless communication [3], image processing [4], and power systems [5]. In recent years, evolutionary computing (EC) [6], such as genetic algorithm [7]–[14], particle swarm optimization (PSO) [15]–[19], differential evolution [20]–[29], estimation of distribution algorithm [30]–[34], and ant colony algorithm [35]–[40], have been widely used to optimize the performance of various global problems [41]–[43]. As a kind of EC algorithm, the

PSO algorithm [44] has been widely used as an effective optimization tool to solve various optimization problems in many real-world applications due to its easy implementation and fast convergence [45]. In the PSO algorithm, the behavior of a flock of birds is simulated, all particles are randomly initialized, and then the global optimal solution is sought based on individual experience and guidance from other particles [46], [47].

In the era of big data, the dimensions of optimization problems are increasing rapidly, and the complexity of optimization problems will also increase exponentially.

Although the PSO algorithm is simple and effective, it faces challenge in dealing with large-scale optimization problems, such as premature convergence and difficulty in parameter control [48]–[50].

As the competitive learning strategy has shown potential in solving large-scale optimization problems [51], this paper proposes a simple and effective method, which is named competition-based stochastic learning particle swarm optimization algorithm (CSLPSO). Different from the competitive learning strategy in existing algorithms [52], in order to increase the diversity of particles and avoid premature convergence, CSLPSO compares the fitness value of one particle with all other particles, and that particle will randomly choose to learn from the other particle that is better than it.

In summary, this paper proposes a CSLPSO, with main contributions summarized as below:

- 1) A competition-based stochastic learning is proposed to improve the exploration ability of PSO.
- 2) A parameter control mechanism suitable for the CSLPSO algorithm is proposed, so that the algorithm does not require manual parameter configuration.

The rest of this article is organized as follows. Section II provides a brief overview of the background and related work. The proposed CSLPSO algorithm is described in detail in Section III. Section IV describes the experimental setup and analysis of the results. Finally, Section V summarizes the paper and discusses future research directions.

II. BACKGROUND AND RELATED WORK

A. Particle Swarm Optimization

PSO stems from the behavioral study of bird predation. Its core idea is to use the sharing of information by individuals in the group to make the movement of the whole group to obtain

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a feasible solution to the problem. In PSO, a particle represents a solution and a group of particles represents a swarm. In a D -dimensional search space, the position of the i^{th} particle of the t^{th} generation is denoted by the vector $x_i(t)=[x_{i,1}(t), x_{i,2}(t), \dots, x_{i,D}(t)]$, and the velocity is denoted by the vector $v_i(t)=[v_{i,1}(t), v_{i,2}(t), \dots, v_{i,D}(t)]$. The formula for updating the velocity and position of particle i is as follows:

$$v_i(t+1) = wv_i(t) + c_1r_1(pbest_i(t) - x_i(t)) + c_2r_2(gbest(t) - x_i(t)) \quad (1)$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (2)$$

where t is the iteration (generation) number; w is termed inertia weight; c_1 and c_2 are the acceleration coefficients; r_1 and r_2 are two vectors randomly generated within $[0,1]$; $pbest_i(t)$ and $gbest(t)$ are the best solution of the i^{th} particle found so far and the best solution found by all particles so far.

The complete algorithm of PSO can be found in **Algorithm1**, which is mainly divided into three steps, which are described as follows:

First, each particle is initialized randomly, the fitness value of each particle is calculated, and the global optimal value is obtained. Then, the velocity and position of the particles are updated according to Eq (1) and (2), and new fitness values are calculated. Finally, the historical optimal position and the historical optimal value of the particle are updated, and the global optimal position and optimal fitness value are updated. The above steps repeat until the stop condition is reached, i.e., t reaches the maximum number of iterations. When the stop condition is reached, the algorithm outputs the optimal solution found and stops running.

B. Related Work on PSO

Many researchers are working to improve the search capabilities of PSO. For example, the PSO-LVIW algorithm [46][46] focuses on the selection of inertia weights, where the updated equation for the inertia weight w at the k^{th} iteration is as follows:

$$w = w_1 - (w_1 - w_2) \times \frac{k}{maxiter} \quad (3)$$

where $maxiter$ represents the maximum number of iterations, and w_1 and w_2 represent the maximum and minimum inertia weights, respectively. In [46], Shi and Eberhart noticed that better performance is achieved if $w_1=0.9$, $w_2=0.4$.

In PSO algorithms, inertia weights are usually used to balance global search and local search [53],[54], where a larger inertia weight is conducive to better global exploration, and a smaller value is beneficial to local search. Inertial weights may lack global search capabilities when they are small. When the problem to be solved is complicated, the PSO-LVIW algorithm may not be able to find the optimal solution required. It is worth noting that, like the inertia weight, the acceleration coefficient also has a special research interest in improving the search ability of the PSO algorithm. In [55], a new PSO variant is proposed, the main contribution of which is to update the PSO acceleration coefficient using the s-type function.

Over the past few years, several PSO variants have been proposed to overcome the limitations of the original PSO algorithm and historical PSO variants. Such as multi-objective optimization [56],[57] and large-scale optimization [58]-[60]. In [52], a competitive group optimizer (CSO) is designed to design a large-scale optimization mechanism. With this pairwise competition mechanism, the particles that lose the competition adjust their position according to the winning particle. In [61], the authors propose a new PSO variant, called Prey-Predator PSO, which implements capture, escape, and reproduction strategies that can help increase convergence speed and reduce computation time.

Algorithm 1 PSO

Input: maximum number of iterations G , the population size N , the particle dimension D

Output: the final solution $fitness_{gbest}$

Begin

1: Initialize velocity v and x for each particle;

2: Evaluate particles and set $pbest$;

3: The smallest fitness value from all $pbest_i$ is selected as $gbest$

4: **For** $t = 1$ to G :

5: **For** $i = 1$ to N :

6: Update the velocity and position of particle i according to (1) and (2);

7: Evaluate particle i ;

8: **If** $fitness(x_i) < fitness(pbest_i)$:

9: $pbest_i = x_i$;

10: **End If**

11: **If** $fitness(pbest_i) < fitness(gbest)$:

12: $gbest = pbest_i$;

13: **End If**

14: **End For**

15: **End For**

End

III. PROPOSED CSLPSO ALGORITHM

PSO algorithm has the advantages of simplicity and fast convergence speed in solving low-dimensional problems, but it often does not work well in high-dimensional problems. So we propose a competition-based stochastic learning strategy.

A. Competition-based Stochastic Learning Strategy

The PSO algorithm learns from $pbest$ and $gbest$ when updating the v . But the optimal fitness value of an individual in a high dimension does not mean that the position of a certain dimension of the individual is optimal, so learning from $gbest$ often causes the algorithm to converge prematurely and fails to achieve good results.

In order to enable individuals to learn from better individuals, we propose a competitive learning strategy that mainly learns only from $pbest$. Before updating, every individual compares the fitness value with all other individuals, and the better individual is stored in a set S . In order to increase the diversity, a random selection manner is used to select better individuals to learn.

To sum up, Eq (1) is rewritten as follows:

$$v_i(t+1) = wv_i(t) + c_1r_1(pbest_{item1}(t) - x_i(t)) + c_2r_2(pbest_{item2}(t) - x_i(t)) \quad (4)$$

where $pbest_{item1}$ and $pbest_{item2}$ represent individuals randomly selected from the set S . Note that $pbest_{item1}$ and $pbest_{item2}$ can be the same.

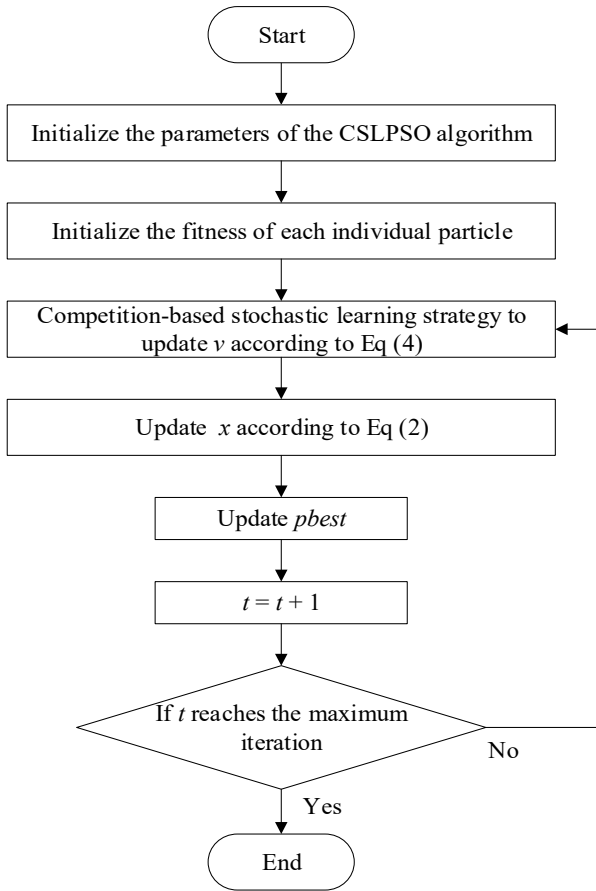


Fig. 1. Flowchart of the CLSPSO algorithm.

B. Parameter Control Mechanism

The effect of the PSO algorithm largely depends on the selection of the inertia parameter w and the acceleration factors c_1 and c_2 , which make the results of the algorithm unstable and increase the difficulty in parameter configuration. Through extensive experiments, we have summarized a mechanism to control the parameters of CSLPSO, as follows:

$$\begin{cases} w = \text{rand}(0.3, 0.5) \\ c_1 = c_2 = 1.4 - w \end{cases} \quad (5)$$

Through this parameter control mechanism, the dependence of the PSO algorithm on manual parameter configuration is avoided, which makes the algorithm simpler to use.

C. The Complete CSLPSO

This section introduces the complete CSLPSO. The complete algorithm is shown in **Algorithm 2** and Fig. 1, the main steps are as follows:

First, according to the competition-based stochastic learning strategy, each particle is compared to other particles and the better particles are put into the set S . Second, each particle needs to randomly select $pbest_{item1}$ and $pbest_{item2}$ when updating v and x for each dimension and update them according to Eq (4) and (2), respectively. This method can make each particle learn from different particles at the same

time, which greatly increases the exploration range of particles, avoids premature convergence of the algorithm, and increases the possibility of finding the global optimal solution. Third, the fitness values of the particles before and after the update are compared, and the better particles are saved. The above steps are repeated until the stopping condition is reached, i.e., t reaches the maximum number of iterations. When the stop condition is reached, the algorithm outputs the optimal solution found and stops running.

Algorithm 2 CSLPSO

Input: maximum number of iterations G , the population size N , the particle dimension D

Output: the final solution $gbest$

Begin

```

1: Population initialization;
2:  $w = \text{rand}(0.3, 0.5)$ ;
3:  $c_1 = c_2 = 1.4 - w$ ;
4: For  $i = 1$  to  $G$ :
5:   For  $j = 1$  to  $N$ :
6:      $S = \emptyset$ ;
7:     For  $u = 1$  to  $N$ :
8:       If the fitness of  $pbest_u \leq$  the fitness of  $pbest_j$ :
9:          $S \leftarrow u$ ;
10:      End If
11:    End For
12:    For  $k = 1$  to  $D$ :
13:      Randomly choose two particle indexes  $item_1, item_2$  from  $S$ ;
14:      Update  $k^{th}$  dimension  $v_i$  and  $x_i$  using Eq (4) and (2), respectively;
15:    End For
16:    Calculate the fitness value of particle  $j$ ;
17:  End For
18:  Update  $pbest$  of particle  $j$ ;
19: End For
End
  
```

IV. EXPERIMENT

A. Experimental Setting

The experiments are carried out on the benchmark functions in Table I, including five unimodal functions and seven multimodal functions. All 12 test functions have no constraints, making it easy to compare the optimization performance. All the functions are of $D = 1000$ dimensions. To obtain more reasonable results, each experiment is run 20 times independently.

For all competitors and functions, the maximum number of iterations G is set to 2,000 and the population size N is set to 100. CSO [52] is a competitive optimizer for solving large-scale problems, and AWPSO [55] is an adaptive particle swarm optimizer. We compared the results of CSLPSO, CSO, and AWPSO. In order to ensure a fair comparison, the ϕ in CSO is taken as 0.1, and the AWPSO uses the original parameter settings in the paper. To enhance clarity, the best results are highlighted in **boldface**.

B. Comparison with State-of-the-Art Algorithm

The comparison results of the 12 benchmark functions are shown in Table II. “+”, “-”, and “=” indicate that the CSLPSO is significantly better, worse, or similarly to the corresponding algorithm obtained according to the Wilcoxon’s rank-sum test at the 0.05 significance level.

As can be seen from Table II, CSLPSO is significantly better than CSO in many functions, except for the significantly worse CSLPSO in f_2 and f_3 . Compared with AWPSON, the performance of CSLPSO is significantly better than that of AWPSON except that there is no significant difference in the performance of the f_7 function. In addition, CSLPSO achieved 9 best results, CSO achieved 2 best results, and AWPSON achieved 1 best result (shown in **boldface**). That is, the CSLPSO achieves the best results on the most problems. In f_7 , although the CSLPSO does not achieve the best results, it is significantly better than the CSO and does not differ significantly from AWPSON, indicating that the overall effect is also good. These demonstrate the high efficiency of CSLPSO.

TABLE I
THE 12 BENCHMARK FUNCTIONS

Name	Function	Range
Sphere	$f_1(x) = \sum_{i=1}^D x_i^2$	$[-100, 100]^D$
Quadric	$f_2(x) = \sum_{i=1}^D \left(\sum_{j=1}^i x_j \right)^2$	$[-100, 100]^D$
Sch's 2.21	$f_3(x) = \max_i (x_i , 1 \leq i \leq D)$	$[-100, 100]^D$
Step	$f_4(x) = \sum_{i=1}^D (\lfloor x_i + 0.5 \rfloor)^2$	$[-100, 100]^D$
Noise	$f_5(x) = \sum_{i=1}^D ix_i^4 + \text{random}[0, 1)$	$[-1.28, 1.28]^D$
Rosenbrock	$f_6(x) = \sum_{i=1}^D [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$	$[-10, 10]^D$
Schwefel	$f_7(x) = \sum_{i=1}^D -x_i \cdot \sin(\sqrt{ x_i }) + 418.9829 \times D$	$[-500, 500]^D$
Rastrigin	$f_8(x) = \sum_{i=1}^D [x_i^2 - 10 \cos(2\pi x_i) + 10]$	$[-5.12, 5.12]^D$
Ackley	$f_9(x) = -20 \exp \left(-0.2 \sqrt{\frac{1}{D} \sum_{i=1}^D x_i^2} \right) - \exp \left(\frac{1}{D} \sum_{i=1}^D \cos 2\pi x_i \right) + 20 + e$	$[-32, 32]^D$
Griewank	$f_{10}(x) = \frac{1}{4000} \sum_{i=1}^D x_i^2 - \prod_{i=1}^D \cos \left(\frac{x_i}{\sqrt{i}} \right) + 1$	$[-600, 600]^D$
Generalized penalized 1	$f_{11}(x) = \frac{\pi}{D} [10 \sin^2(\pi y_1) + \sum_{i=1}^{D-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1}) + (y_D - 1)^2] + \sum_{i=1}^D u(x_i, 10, 100, 4)]$ where $y_i = 1 + \frac{1}{4}(x_i + 1), u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m, & x_i > a \\ 0, & -a \leq x_i \leq a \\ k(-x_i - a)^m, & x_i < -a \end{cases}$	$[-50, 50]^D$
Generalized penalized 2	$f_{12}(x) = 0.1 \left\{ \sin^2(3\pi x_1) + \sum_{i=1}^{D-1} (x_i - 1)^2 [1 + \sin^2(3\pi x_{i+1})] + (x_D - 1)^2 [1 + \sin^2(2\pi x_D)] \right\} + \sum_{i=1}^D u(x_i, 5, 100, 4)$	$[-50, 50]^D$

For f_2 , the influence of different dimensions of x^d on the overall fitness value is different, and the lower the dimension of x^d , the greater the influence on the fitness value, and this influence will increase with the increase of dimension. The social behavior of CSO ($\bar{X}_k$) can reduce this effect to a certain extent, so that the mean value obtained by CSO is better than that of CSLPSO. CSLPSO does not eliminate this effect, but greatly increases the search range of particles through the CSL strategy. Although this method will make the variance of the obtained solution larger, and the average value

is worse than that of CSO, the optimal value of multiple solutions obtained by running CSLPSO multiple times is even better than that of CSO to obtain the optimal solution, that is, the lower bound of CSLPSO is better than the lower bound of CSO.

For f_3 , the fitness value of the function depends on the largest absolute value of each dimension $|x^d|$, and the social behavior of the CSO ($\bar{X}_k$) can drive particle search in the optimal direction, while the CSLPSO does not guarantee that the particle will search in the optimal direction.

In summary, the performance of CSLPSO is better than that of CSO and AWPSON on most functions.

TABLE II
RESULTS ON THE 12 BENCHMARK FUNCTIONS

Function	CSLPSO		CSO		AWPSON	
	mean	std	mean	std	mean	std
f_1	3.38E-01	2.78E-01	3.33E+02(+)	3.35E+01	1.70E+04(+)	9.75E+02
f_2	1.25E+05	4.61E+04	9.02E+04(-)	7.84E+03	1.17E+06(+)	1.94E+05
f_3	1.16E+01	4.48E-01	9.05E+00(-)	3.01E-01	3.86E+01(+)	1.11E+00
f_4	4.45E+02	9.14E+01	1.20E+03(+)	1.15E+02	1.91E+04(+)	1.48E+03
f_5	7.54E-01	9.24E-02	1.04E+00(+)	7.22E-02	4.27E+01(+)	4.43E+00
f_6	1.01E+03	2.65E+01	1.80E+03(+)	7.72E+01	5.63E+04(+)	6.21E+03
f_7	2.79E+05	8.30E+04	3.60E+05(+)	5.81E+03	2.58E+05(=)	9.01E+03
f_8	1.01E+02	9.19E+00	1.47E+02(+)	1.88E+01	3.51E+03(+)	1.46E+02
f_9	2.09E+00	1.62E-01	2.31E+00(+)	7.96E-02	8.22E+00(+)	3.15E-01
f_{10}	9.16E-02	1.26E-01	4.30E+00(+)	5.28E-01	5.30E+02(+)	3.63E+01
f_{11}	3.71E-03	3.72E-03	1.02E-01(+)	1.94E-02	1.64E+04(+)	1.14E+04
f_{12}	1.74E+01	8.17E+00	1.29E+02(+)	1.67E+01	8.90E+05(+)	2.33E+05
Total (+/-/=)	10/2/0		11/0/1			

TABLE III
COMPARE THE RESULTS OF CSLPSO, CSLPSO-W/O-C AND PSO

Function	CSLPSO		CSLPSO-w/o-c		PSO	
	mean	std	mean	std	mean	std
f_1	3.38E-01	2.78E-01	3.00E+04(+)	1.15E+03	1.46E+04(+)	7.44E+02
f_2	1.25E+05	4.61E+04	6.55E+05(+)	7.08E+04	1.39E+06(+)	2.41E+05
f_3	1.16E+01	4.48E-01	1.92E+01(+)	3.93E-01	3.95E+01(+)	1.01E+00
f_4	4.45E+02	9.14E+01	2.96E+04(+)	1.13E+03	1.39E+04(+)	1.37E+03
f_5	7.54E-01	9.24E-02	3.38E+01(+)	2.64E+00	3.22E+01(+)	2.33E+00
f_6	1.01E+03	2.65E+01	5.43E+04(+)	2.50E+03	4.76E+04(+)	3.62E+03
f_7	2.79E+05	8.30E+04	2.71E+05(=)	2.65E+04	2.50E+05(=)	1.08E+04
f_8	1.01E+02	9.19E+00	7.45E+03(+)	3.14E+02	3.60E+03(+)	1.07E+02
f_9	2.09E+00	1.62E-01	7.62E+00(+)	8.81E-02	7.31E+00(+)	6.77E-01
f_{10}	9.16E-02	1.26E-01	2.73E+02(+)	8.55E+00	1.36E+02(+)	8.76E+00
f_{11}	3.71E-03	3.72E-03	1.41E+01(+)	1.26E+00	2.01E+04(+)	9.25E+03
f_{12}	1.74E+01	8.17E+00	1.00E+05(+)	1.81E+04	7.90E+05(+)	1.94E+05
Total (+/-/=)	11/0/1		11/0/1			

Note that CSLPSO-w/o-c does not adopt the parameter control mechanism. CSLPSO-w/o-c and PSO use the same w and c_1, c_2 . w decreases linearly from 0.9 to 0.4, $c_1 = c_2 = 1.9$.

C. Component Analysis of the CSLPSO

It can be seen from Table III that CSLPSO is significantly superior to PSO in other functions, except that there is no significant difference between CSLPSO and PSO in f_7 . In addition, the CSLPSO achieved 11 best results and the PSO achieved 1 best result, i.e., the CSLPSO achieved the best results on the most problems, and in f_7 , although the CSLPSO does not achieve the best results, there is no significant difference in the PSO, indicating that the overall effect is also good. These indicate the effectiveness of the CSL strategy and show that the performance of CSLPSO is better than that of PSO on the whole.

D. Parameter Study of the CSLPSO

The inertia weight and acceleration factor have a great influence on the search ability of the PSO, and the CSLPSO retains these parameters of the PSO, and we have tested the results of different parameters in 12 functions, as detailed in Tables IV, V, and VI.

TABLE IV
RESULTS OF DIFFERENT PARAMETERS ON THE 12 BENCHMARK
FUNCTIONS, WHERE $C_1, C_2=1.1$

Function	$w=0.3$		$w=0.4$		$w=0.5$	
	mean	std	mean	std	mean	std
f_1	2.72E+00	4.36E+00	2.29E+02	3.34E+01	2.25E+03	2.30E+02
f_2	1.31E+05	3.89E+04	2.62E+05	9.28E+04	2.37E+05	4.71E+04
f_3	1.18E+01	3.78E-01	1.28E+01	4.62E-01	1.48E+01	6.79E-01
f_4	4.82E+02	7.87E+01	1.68E+03	2.84E+02	5.27E+03	2.58E+02
f_5	7.34E-01	3.05E-02	2.07E+00	9.86E-02	5.60E+00	5.21E-01
f_6	1.04E+03	2.34E+01	2.05E+03	1.68E+02	5.71E+03	4.22E+02
f_7	2.53E+05	7.10E+04	3.35E+05	9.12E+04	2.89E+05	8.01E+04
f_8	9.53E+01	6.91E+00	2.00E+02	1.45E+01	8.51E+02	6.64E+01
f_9	2.12E+00	4.61E-02	3.02E+00	9.72E-02	4.39E+00	1.69E-01
f_{10}	1.52E-01	2.68E-01	2.77E+00	4.23E-01	2.18E+01	2.38E+00
f_{11}	2.89E-03	2.63E-03	1.61E-01	5.01E-02	9.16E-01	1.10E-01
f_{12}	8.15E+00	2.64E+00	3.47E+02	4.80E+01	7.16E+02	8.15E+01

TABLE V
RESULTS OF DIFFERENT PARAMETERS ON THE 12 BENCHMARK
FUNCTIONS, WHERE $C_1, C_2=1.0$

Function	$w=0.3$		$w=0.4$		$w=0.5$	
	mean	std	mean	std	mean	std
f_1	3.38E+03	2.60E+02	2.58E-01	3.85E-01	2.13E+02	2.58E+01
f_2	1.29E+05	7.37E+03	9.98E+04	1.17E+04	2.27E+05	1.11E+05
f_3	1.13E+01	2.17E-01	1.14E+01	1.70E-01	1.31E+01	6.72E-01
f_4	4.30E+03	2.71E+02	5.04E+02	5.19E+01	1.75E+03	1.66E+02
f_5	3.24E+00	3.07E-01	9.11E-01	7.53E-02	2.21E+00	2.36E-01
f_6	5.98E+03	2.93E+02	1.04E+03	4.99E+01	1.98E+03	1.26E+02
f_7	3.11E+05	7.03E+03	2.24E+05	2.03E+04	2.90E+05	1.03E+05
f_8	1.25E+03	5.56E+01	1.06E+02	1.01E+01	2.38E+02	2.35E+01
f_9	3.58E+00	7.37E-02	2.05E+00	1.24E-01	3.16E+00	1.55E-01
f_{10}	3.13E+01	2.97E+00	1.03E-01	1.57E-01	3.00E+00	4.40E-01
f_{11}	9.95E-01	9.44E-02	1.32E-03	1.68E-03	2.45E-01	4.54E-02
f_{12}	4.92E+02	1.98E+01	1.94E+01	2.58E+00	3.71E+02	5.45E+01

TABLE VI
RESULTS OF DIFFERENT PARAMETERS ON THE 12 BENCHMARK
FUNCTIONS, WHERE $C_1, C_2=0.9$

Function	$w=0.3$		$w=0.4$		$w=0.5$	
	mean	std	mean	std	mean	std
f_1	1.12E+04	8.36E+02	3.70E+03	2.55E+02	5.94E-01	8.71E-01
f_2	2.63E+05	3.71E+04	1.87E+05	3.82E+04	1.09E+05	2.14E+04
f_3	1.17E+01	6.39E-01	1.10E+01	5.64E-01	1.10E+01	2.99E-01
f_4	1.11E+04	6.21E+02	5.13E+03	4.15E+02	6.34E+02	1.49E+02
f_5	6.28E+00	4.82E-01	3.37E+00	2.85E-01	1.13E+00	2.22E-01
f_6	1.50E+04	6.25E+02	6.31E+03	7.22E+02	1.02E+03	3.82E+01
f_7	3.62E+05	5.23E+03	3.66E+05	4.73E+04	2.16E+05	1.39E+04
f_8	4.18E+03	1.18E+02	1.55E+03	6.82E+01	9.86E+01	6.63E+00
f_9	5.33E+00	9.31E-02	3.88E+00	1.13E-01	2.20E+00	1.15E-01
f_{10}	9.89E+01	5.44E+00	3.77E+01	2.76E+00	1.55E-02	1.62E-02
f_{11}	3.04E+00	1.69E-01	1.23E+00	8.75E-02	8.53E-03	5.10E-03
f_{12}	7.34E+02	2.84E+01	5.30E+02	3.86E+01	3.66E+01	1.23E+01

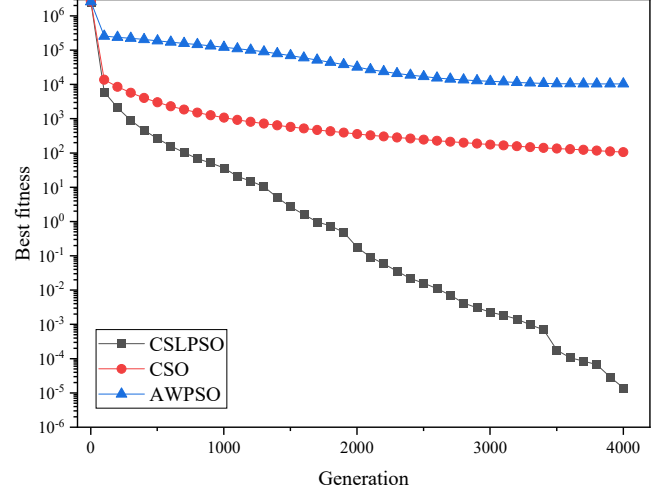
As can be seen from Table IV, V, and VI, $w=0.3$, $c_1=c_2=1.1$; $w=0.4$, $c_1=c_2=1.0$; $w=0.5$, $c_1=c_2=0.9$ perform significantly better than other combinations in 11 functions except f_3 (shown in **boldface**).

As can be seen from Table III, the CSLPSO with the parameter control mechanism is significantly better than the CSLPSO without this method in 12 functions, i.e., CSLPSO is

significantly better than CSLPSO-w/o-c. Moreover, there is no significant difference between the optimal parameter combinations in Table IV, V, and VI and the results obtained by the parameter control mechanism, which further proves the effectiveness of the parameter control mechanism.

E. Algorithm Performance Evaluation

In order to fully analyze the experimental results, we evaluate the performance of the algorithm by examining the optimal fitness obtained in each generation on different functions, see Fig. 2 and Fig. 3.



(a) The curve of best fitness of CSLPSO, CSO, and AWPSO on f_1

Fig. 2. The best fitness of CSLPSO, CSO, and AWPSO in the 1st benchmark problem

On some functions, such as f_1 , the optimal fitness obtained by CSLPSO is significantly better than that obtained by CSO and AWPSO, see Fig. 2. On the whole, CSLPSO has the fastest convergence rate, CSLPSO can still maintain stable convergence, and there is no decline in the convergence rate like the other two algorithms.

In some complex functions, such as f_2 and f_7 , the values of x^d in different dimensions in f_2 will affect the overall fitness; in f_7 , the existence of periodic functions such as $\sin(x)$ leads to many local optimal points in the function, which increases the difficulty of function optimization. Although CSLPSO does not achieve the best results on these functions, it is not very different from the algorithms that achieve the best results.

Combining the results of Table III and Fig. 3, it is not difficult to see that CSLPSO can maintain a stable convergence rate on most functions, even if the performance on some functions is not ideal, it is not the worst one. This shows the effectiveness of the CSLPSO algorithm.

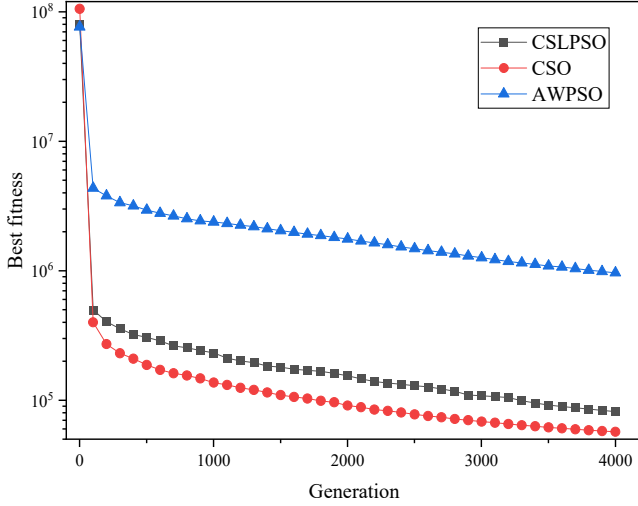
V. CONCLUSION

To solve large-scale optimization problems efficiently, this paper proposes a simple and effective CSLPSO algorithm for solving large-scale problems.

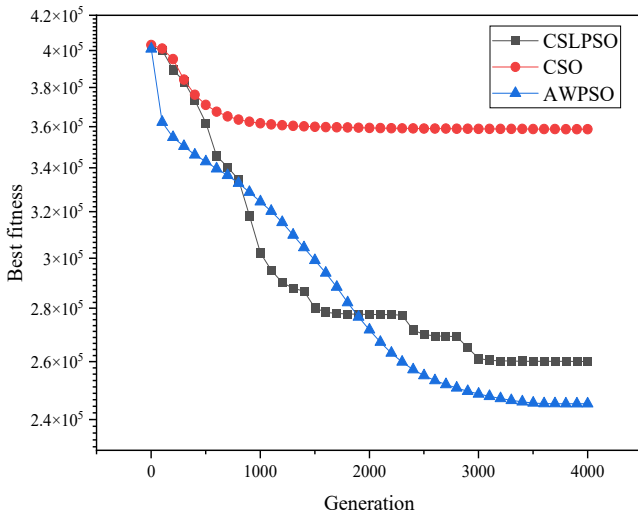
In the algorithm design, we propose a CSL strategy, which improves the exploration ability of particles. At the same time, we also propose a parameter control mechanism for CSLPSO

algorithm, which makes the algorithm not overly dependent on parameter selection like PSO, and the effect is more stable.

Through the observation of the current research, in future work, we will consider the enhanced learning strategy to make the algorithm perform better on the multimodal function.



(a) The curve of best fitness of CSLPSO, CSO and AWPSO on f_2



(b) The curve of best fitness of CSLPSO, CSO, and AWPSO on f_7

Fig. 3. The best fitness of CSLPSO, CSO, and AWPSO in the 2nd, and 7th benchmark problem

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